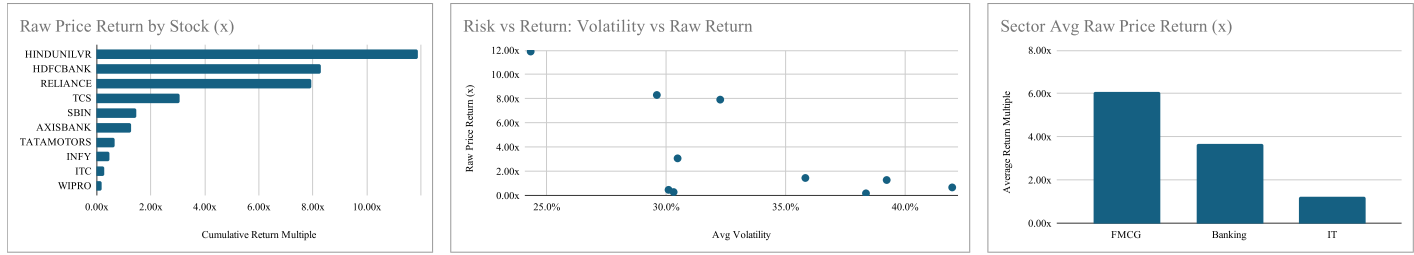


Stock Market Performance Analysis — Summary

10 NIFTY 50 Stocks | January 2000 to April 2021 | 44,219 rows | Built in Excel / Google Sheets with ChatGPT assistance

Executive Summary Chart

Best Performer HINDUNILVR 11.90x return	Highest Return 11.90x ▲ 10.90 vs 1x	Worst Performer WIPRO 0.18x return	Worst Day Loss -39.5% WIPRO	Best Day Gain +27.7% SBIN	Most Traded TATAMOTORS 53.7B shares
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Return Leaderboard and Risk Profile

Stock	Raw Price Return	Avg Daily Return	Avg Volatility	Total Volume	Risk / Return Comment	Stock	Raw Price Return	Stock	Avg Volatility	Raw Price Return
HINDUNILVR	11.90x	+0.09%	+24.33%	7.3B	Wealth creator	HINDUNILVR	11.90x	HINDUNILVR	24.3%	11.90x
HDFCBANK	8.31x	+0.08%	+29.61%	11.2B	Wealth creator	HDFCBANK	8.31x	HDFCBANK	29.6%	8.31x
RELIANCE	7.92x	+0.07%	+32.26%	29.6B	Wealth creator	RELIANCE	7.92x	RELIANCE	32.3%	7.92x
TCS	3.07x	+0.06%	+30.48%	6.9B	Wealth creator	TCS	3.07x	TCS	30.5%	3.07x
SBIN	1.45x	+0.06%	+35.83%	53.3B	Wealth creator	SBIN	1.45x	SBIN	35.8%	1.45x
AXISBANK	1.28x	+0.07%	+39.23%	23.5B	Wealth creator	AXISBANK	1.28x	AXISBANK	39.2%	1.28x
TATAMOTORS	0.67x	+0.05%	+41.97%	53.7B	Below 1x; destroyed raw price value	TATAMOTORS	0.67x	TATAMOTORS	42.0%	0.67x
INFY	0.47x	+0.01%	+30.10%	11.0B	Below 1x; destroyed raw price value	INFY	0.47x	INFY	30.1%	0.47x
ITC	0.29x	+0.03%	+30.31%	38.1B	Below 1x; destroyed raw price value	ITC	0.29x	ITC	30.3%	0.29x
WIPRO	0.18x	+0.02%	+38.36%	11.9B	Below 1x; destroyed raw price value	WIPRO	0.18x	WIPRO	38.4%	0.18x
						Sector	Avg Raw Price Return			
						FMCG	6.09x			
						Banking	3.68x			
						IT	1.24x			

Project Overview

Title [1]	Stock Market Performance Analysis — NIFTY 50
Dataset	10 NIFTY 50 Stocks January 2000 to April 2021 44,219 rows
Tool Used	Microsoft Excel / Google Sheets
AI Assistance	ChatGPT
Data Note	Cumulative returns use raw close/prev-close price movement and are not split-adjusted total returns.

Data Cleaning Steps

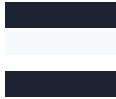
- Removed the Trades column — 19,679 null values, not required for analysis.
- Converted Date column to proper date format.
- Added Turnover by Cr column — Turnover divided by 10,000,000.
- Added Price Check column — flagged rows where Low > High as Error.
- Handled Deliverable Volume nulls — 2,565 rows left blank as unavailable in source.
- Added Year Month for pivot grouping and Year for slicer filtering.
- Identified 18 split / corporate-action events with extreme daily returns between -49% and -93%; documented in dashboard.

Financial Calculations

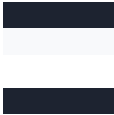
Metric	Formula	Interpretation
Daily Return %	$(\text{Close} - \text{Previous Close}) / \text{Previous Close}$	Measures daily percentage price movement.
Cumulative Return	$\text{Previous Cumulative} \times (1 + \text{Daily Return})$	Measures total investment growth from the first trading day.
Moving Averages	Average of last 10, 20, or 50 closing prices	Smooths daily noise to show trend direction.
Volatility	20-day rolling standard deviation of daily returns	Measures price movement risk; higher volatility = higher risk.
Correlation	CORREL on daily returns of two stocks	Measures how similarly two stocks move; range is -1 to +1.
Sharpe Ratio	$(\text{Average Daily Return} / \text{Std Dev of Returns}) \times \sqrt{252}$	Measures return earned per unit of risk.

Key Findings and Trends

Topic [2]	Summary
Best Performing Stock — HINDUNILVR	Delivered 11.90x raw price return with the lowest average volatility of 24.3%; it behaved like a defensive FMCG compounder through major crises.
Worst Performing Stock — WIPRO	Delivered only 0.18x, meaning ₹1 invested fell to ₹0.18 by 2021 in raw price terms.
Four Stocks Lost Money	WIPRO, ITC, INFY, and TATAMOTORS ended below 1x, highlighting stock-selection risk.
Risk vs Return	Higher volatility did not translate into higher reward: TATAMOTORS had the highest volatility at ~42% but only 0.67x return.
COVID Crisis Behaviour	TATAMOTORS fell sharply during the crash but recovered fastest; HINDUNILVR protected capital as the defensive outlier.
2008 Global Financial Crisis	2008 was the worst year for most stocks; HINDUNILVR was the only positive-return defensive stock while TATAMOTORS rebounded hardest in 2009.
Correlation Findings	AXISBANK/SBIN had the highest correlation at 0.4615, TCS/WIPRO had 0.4152, and INFY/ITC was the best diversification pair at 0.0832.
Sector Summary	FMCG averaged ~6.09x, Banking ~3.68x, and IT ~1.24x; company-specific quality mattered more than sector alone.
Trend Takeaway	Post-crisis recoveries were explosive, but long-term compounding favored quality defensive stocks over volatile cyclical.



Stock Market Performance Analysis — Summary	
10 NIFTY 50 Stocks January 2000 to April 2021 44,219 rows Built in Excel / Google Sheets with ChatGPT assistance	
Use of ChatGPT in This Project	
Step	How ChatGPT Helped
1 — Data Cleaning Guidance	Suggested removing Trades, converting dates, and adding Turnover by Cr and Price Check.
2 — Daily Return Formula	Provided IF logic to avoid calculating returns across different symbols.
3 — Moving Average Formula	Explained stock-specific MA10, MA20, and MAS0 logic for stacked data.
4 — Volatility Explanation	Defined volatility and rolling STDEV in simple investment-banking terms.
5 — Correlation Analysis	Explained CORREL setup and diversification interpretation for a 10x10 matrix.
6 — Dashboard Design	Recommended price, moving average, return comparison, and pivot summary visuals.
7 — Moving Average Crossover	Explained Golden Cross and Death Cross patterns for dashboard commentary.
8 — Performance Summary Writing	Helped structure best/worst performers, risk-return, and sector conclusions.
Conclusion	
The analysis shows that long-term patient investing in defensive quality stocks significantly outperformed chasing high-risk cyclicals. HINDUNILVR combined 11.90x raw price return with the lowest risk profile, while 4 of 10 stocks destroyed investor wealth in raw price terms. Correlation anal diversification should combine banking or cyclical exposure with FMCG defensives like HINDUNILVR to reduce portfolio risk without sacrificing return potential.	



ysis suggests

[1] Source: User-provided project summary and existing workbook dashboard.

[2] Source: User-provided project narrative; metrics cross-checked against Dashboard Helper where available.